

Consultation analysis: a production architecture

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One central service that turns a department's consultation responses into signed-off themes and a filterable dashboard.

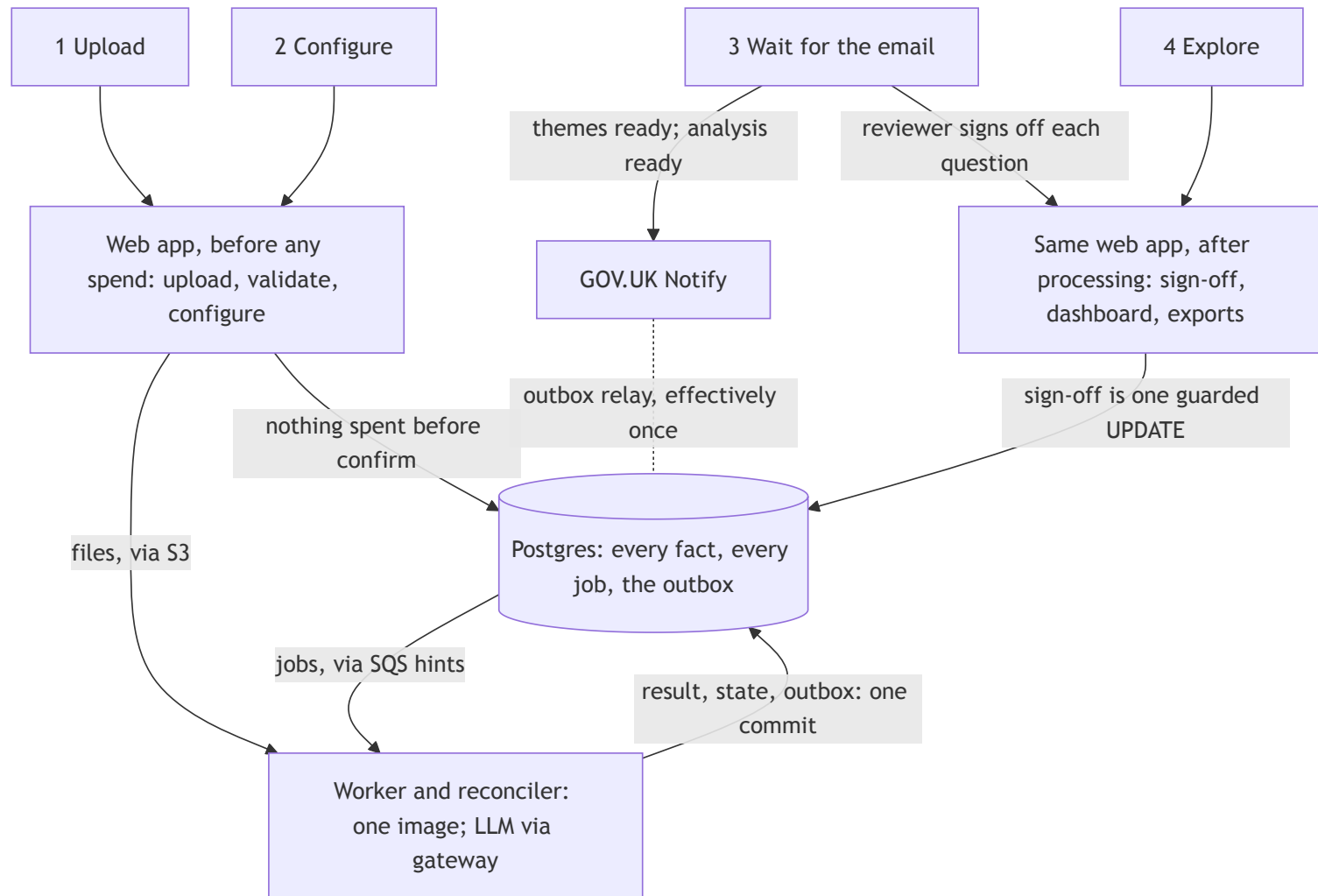


Diagram 1. One database holds every fact and job, reviewers sign off before tagging, and each email is written in the commit that earns it.

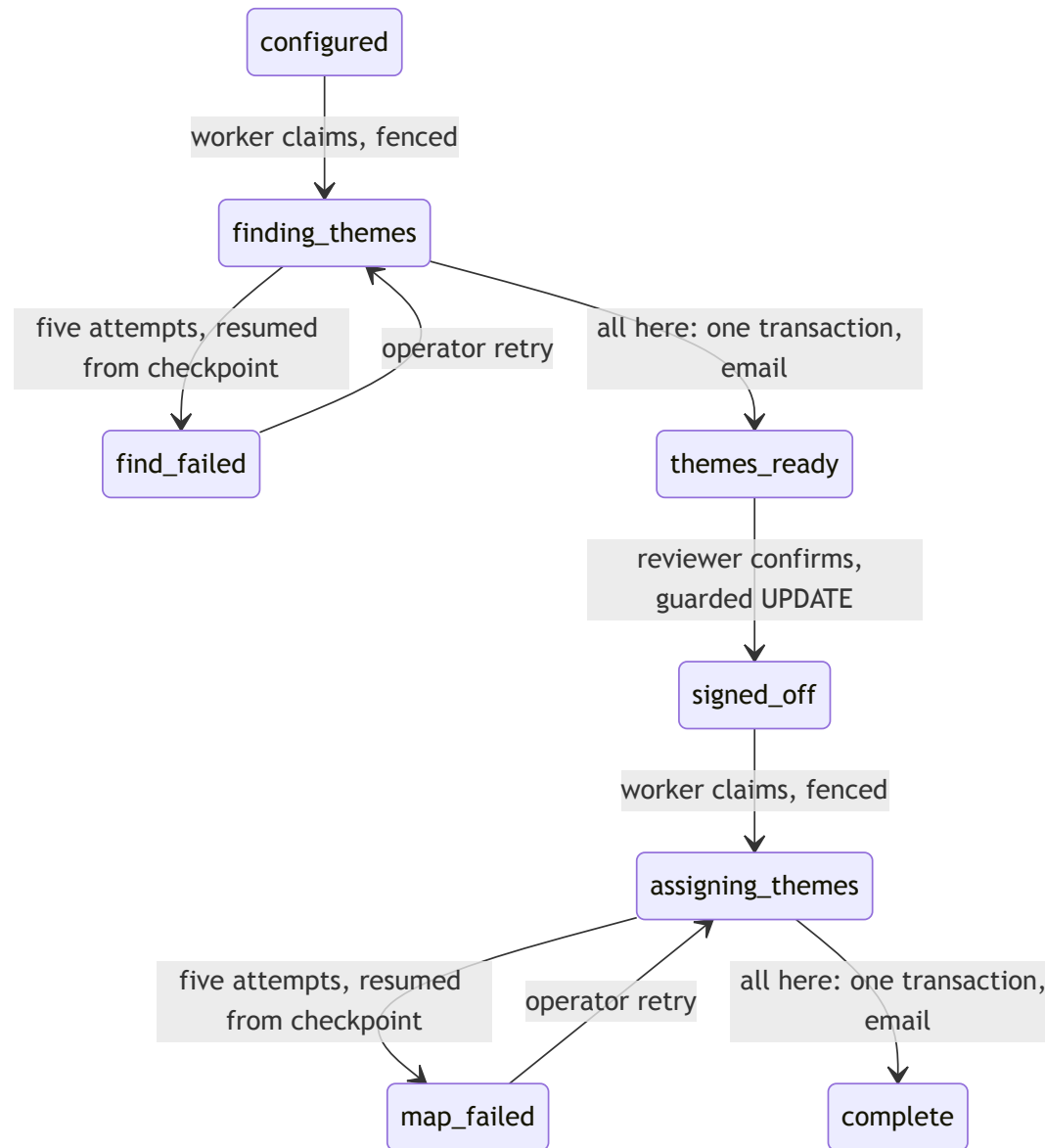


Diagram 2. Each question runs, fails and retries alone; awaiting_review, ready and both emails come from one guarded UPDATE in the commit finishing a question.

The design, in bullets

- A named reviewer signs off each question's themes; every edit is kept. Uploads are validated and costed before spend; no demographic reaches a prompt. One platform serves every department.
- I've added sign-off per question before tagging: in i.AI's published evaluation reviewers left roughly three mappings in four unchanged and corrected the rest. Confirm-as-is is one click.
- Postgres holds every fact and all pipeline state; S3 is archive only. The job table is the truth and queue messages are hints, so a duplicate or lost message is harmless.
- One job per open question with a lease, a fencing token and per-batch checkpoints: a dead worker's job resumes from its checkpoint; a zombie can't write on a stale fence.
- The fan-in is one transaction behind a row lock: mark the question done, flip the consultation if all are, queue the email. One finisher wins; the reconciler covers a crash.
- One answer row per respondent, question and option; a respondent snapshot makes 'villagers who cycle to work and oppose' one indexed query; nothing is edited in place, so counts reproduce.
- Responses are untrusted input: no tools, answers sent as JSON data, labels an enum of theme keys, small batches to bound the blast radius. Containment first; detection is a bonus.
- Encrypted at rest in London. Only the question, the open answer and any linked closed answer reach the model, via a gateway logging each job's inference region for the DPIA.
- Parity first, the manual process alongside and pilot consultations replayed as the gate. Then sign-off and dashboard, then self-serve behind department caps. A small team, two quarters.
- Model spend is small beside reviewer time, as the water commission press release shows; throughput and sign-off time bind. Cost is a guard: an estimate at Confirm, a department budget.
- Least sure, so I'd test first: the dashboard's query plan under a selective filter on the largest consultations, and whether per-question sign-off is bearable at dozens of questions.

I read i.AI's public repository and decision records first; where 'every department' changes the problem I've gone further.

I used Claude throughout, to research, to stress-test alternatives and to draft the diagrams. The choices are mine.